



Steps of VaR calculation, portfolio VaR

Risk Management in Financial Institutions

Lecture 2

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The Value at Risk (VaR) - *reminder*

○ Value at Risk, V@R

- maximum future loss V_t , on a portfolio with time value t
- after a given time horizon τ (e.g.: 1 day, 10 days,...)
- for a given α confidence level (e.g.: 95%, 99%, ...)
- assuming a normal course of business

„What could be the maximum loss possible tomorrow with a 99% probability?”

$$\text{Prob}[(V_{t+\tau}) > (V_t + \text{Var}(V_t))] = \alpha$$

If the **distribution function** of $V_{t+\tau}$ v.v., F , and its **inverse**, F^{-1} , are known, then

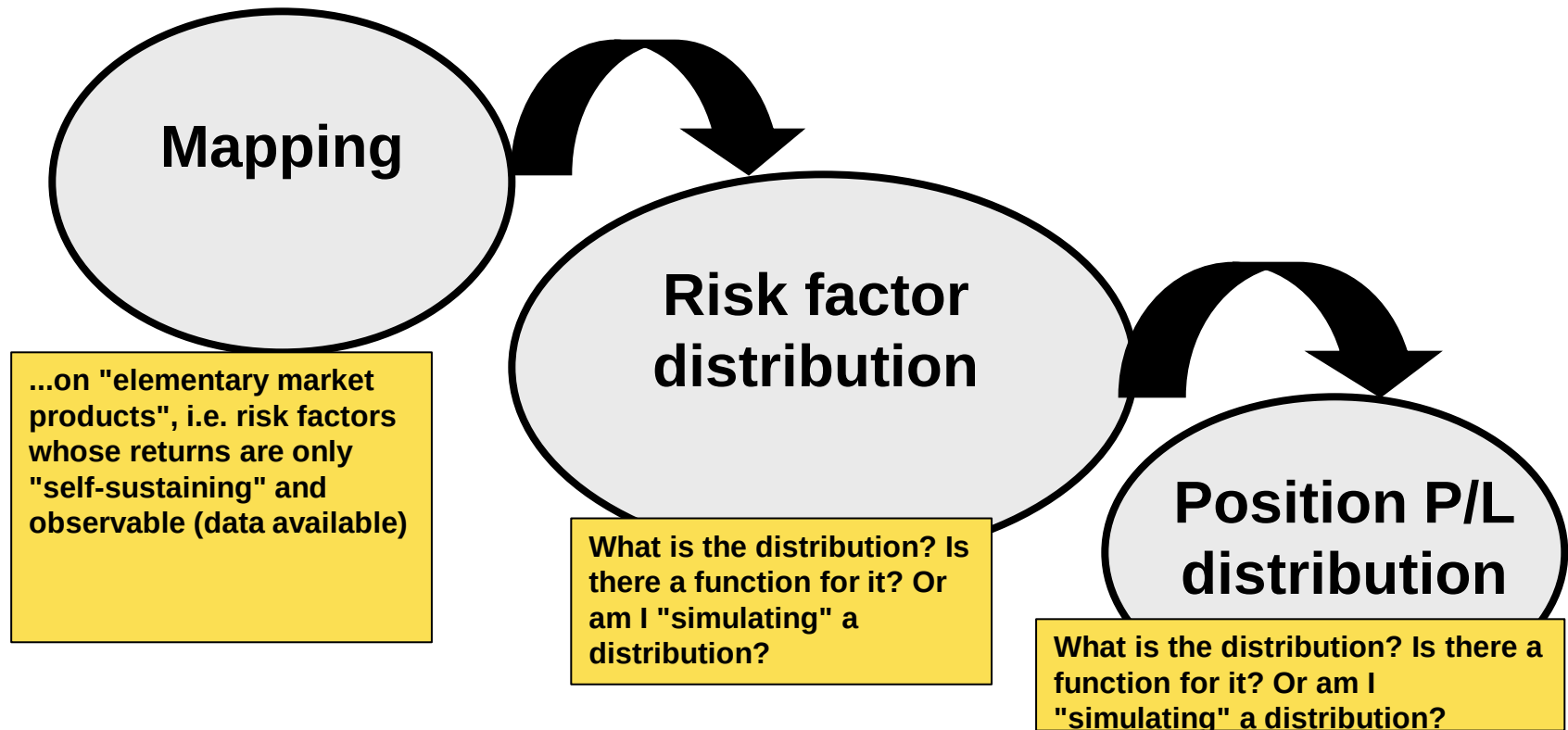
$$F((V_{t+\tau} - V_t) > \text{Var}(V_t)) = \alpha$$

$$F(dV_\tau < \text{VaR}) = 1 - \alpha$$

$$\text{VaR}(V) = F^{-1}(dV_\tau, 1 - \alpha)$$

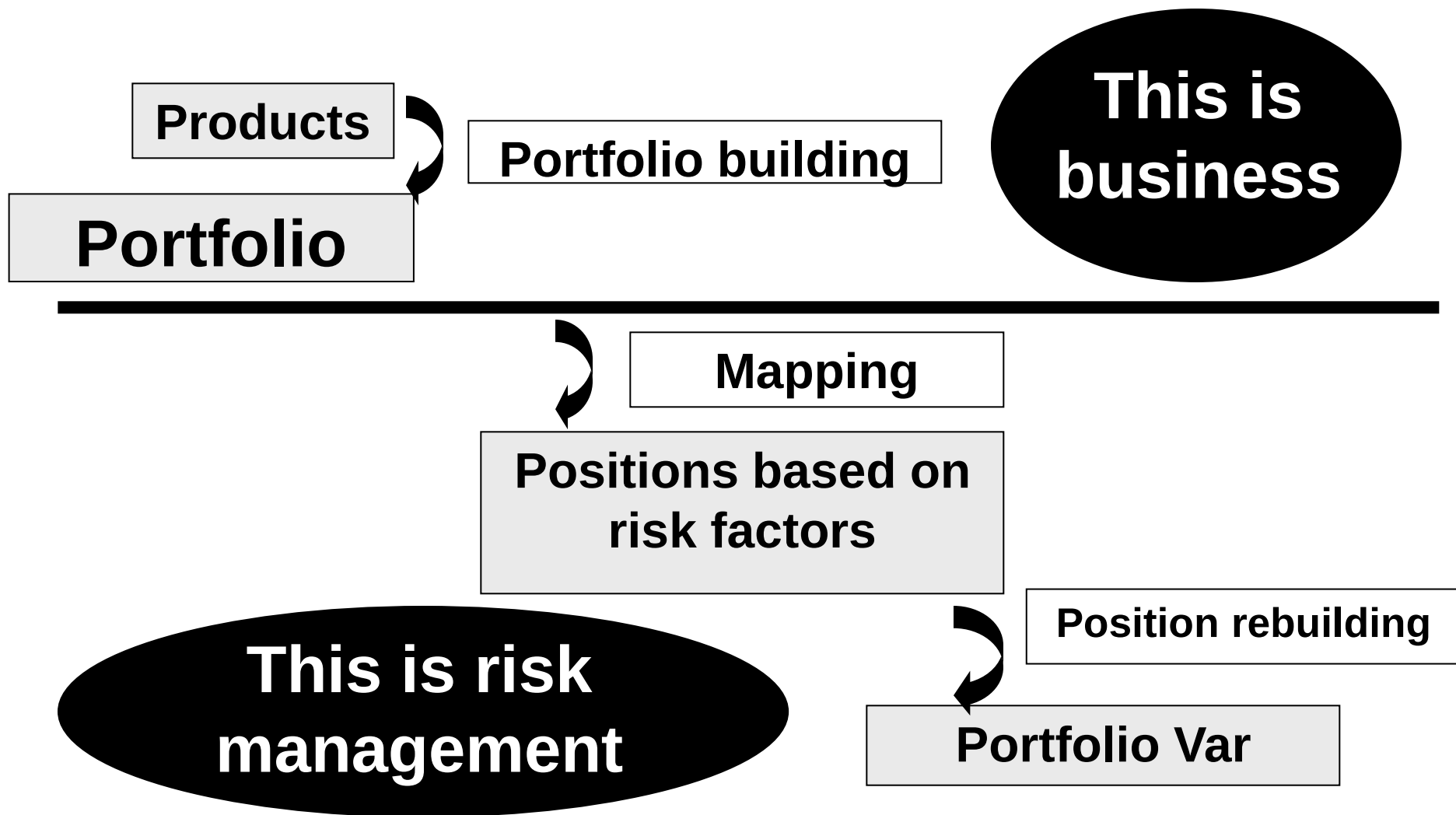
*In other words, VaR is the inverse of the **distribution function of the change in portfolio value ("return") over time τ** , taken at $1 - \alpha$.*

Steps of portfolio VaR calculation - disassembling and reassembling the portfolio



Mapping the position: the key to VAR is tracing back to the risk factor

The "repackaging"



The key questions

Mapping: is the change in value a linear or non-linear function of the return on risk factors?

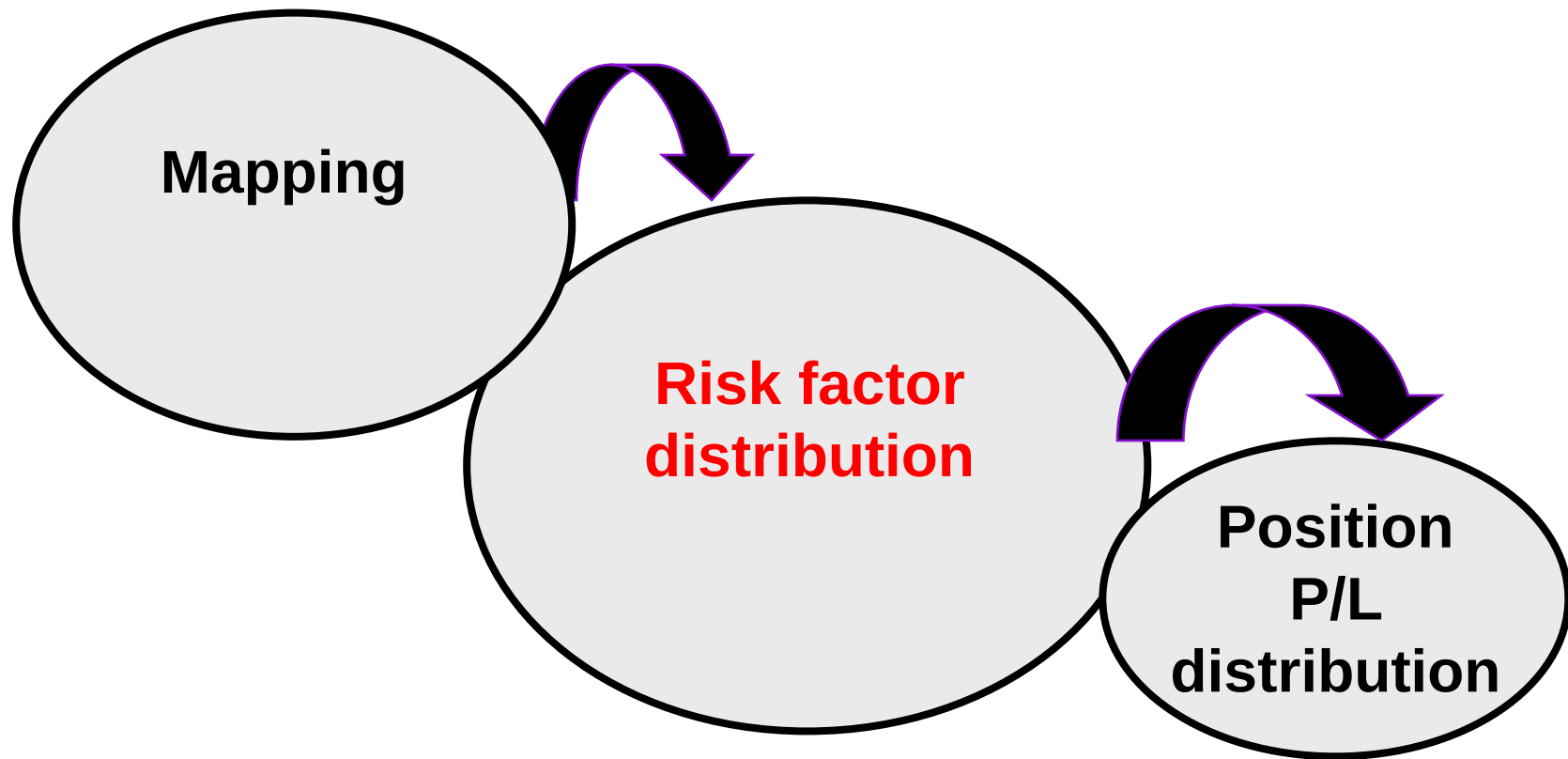


Statistical properties of risk factor returns: do we know the type of distribution?

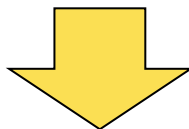


Distribution of value changes: taking into account the co-movement of risk factors

Logical model for determining the Value at Risk



Methods for VaR determination



different methods/assumptions for determining distributions

- Analytic or delta normal method (also known as parametric/variance-covariance method)
- Historical method
- Simulation
 - Historical simulation (bootstrapping) sampling from historical data
 - Monte Carlo Simulation from known or estimated distribution

Assumption on the distribution in delta normal method

- Assume that the distribution of returns is **known and normal!**
- Then the value at risk depends only on the expected value, the variance (covariance) and the quintiles of the standard normal distribution!

VaR easily, quickly - 2. step

Delta normal VaR of a portfolio with N identical shares

- We know that the **risk factor** of a share is its **return**:

$$dV = N \cdot dS = N \cdot S \cdot dr$$

where $V=N \times S$, is the value of the portfolio; N : number of units, S : exchange rate, r : logarithmic return

- If we assume the **normality** of the **share return**, then we only need to estimate the **expected value** and the **standard deviation** of the **return**...

$$r \sim N(\mu_r, \sigma_r) \Rightarrow V \cdot r \sim N(V \cdot \mu_r, V \cdot \sigma_r)$$

...because then VaR depends essentially on the standard deviation, since...

$$P(\Delta V < VaR) = P(V \cdot r < VaR) = (1 - \alpha)\%$$

... and VaR expressed from it using the inverse of the normal distribution:

$$VaR = \mu + \sigma N^{-1}(X)$$

VaR easily, quickly - 2. step

If the short-term expected return is 0, then the delta normal VaR depends only on the standard deviation!

$$\text{VaR} = \text{Multiplier} \cdot \text{Position size} \cdot \text{Standard deviation}$$

- **Multiplier** is a parameter that depends on the choice of risk level.

$$N^{-1}(1-\alpha)$$

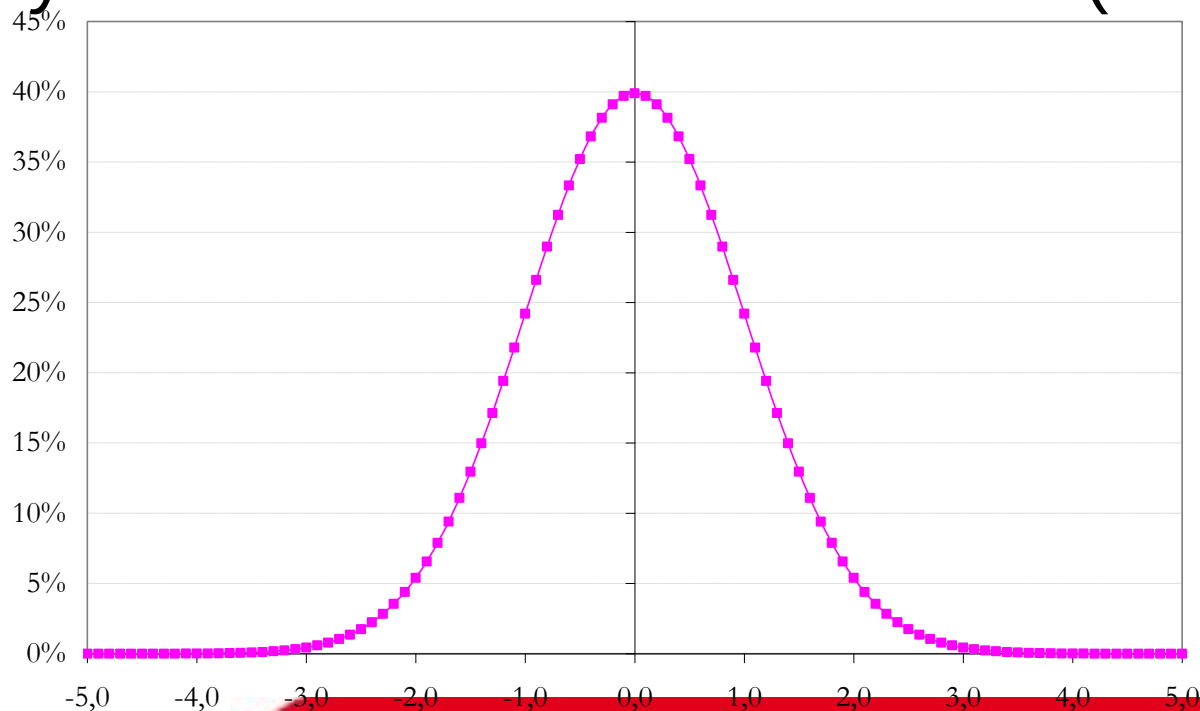
- $\alpha = 99\%$ Multiplier = -2,326

- $\alpha = 95\%$ Multiplier = -1,645

- Quick, calculable in your head..
- ..but gives somewhat "peaceful", low loss estimates (max loss is a little more than two times the standard deviation...)

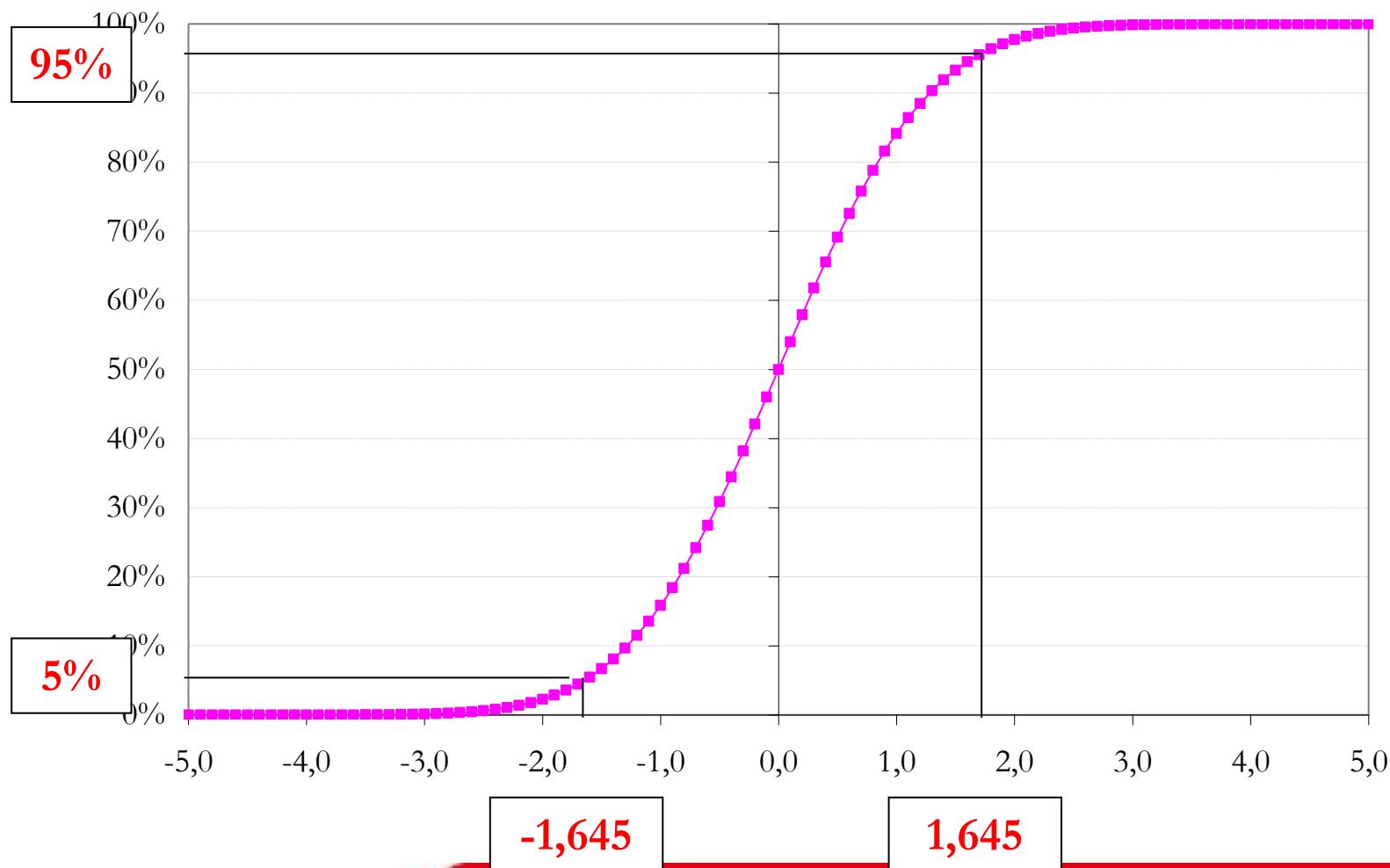
Normal distribution

- **Normal distribution:** continuous distribution
- Characteristics: mean, standard deviation: $N(\mu, \sigma)$
- Standard normal distribution: $N(0, 1)$
- Density function of normal distribution (bell curve)



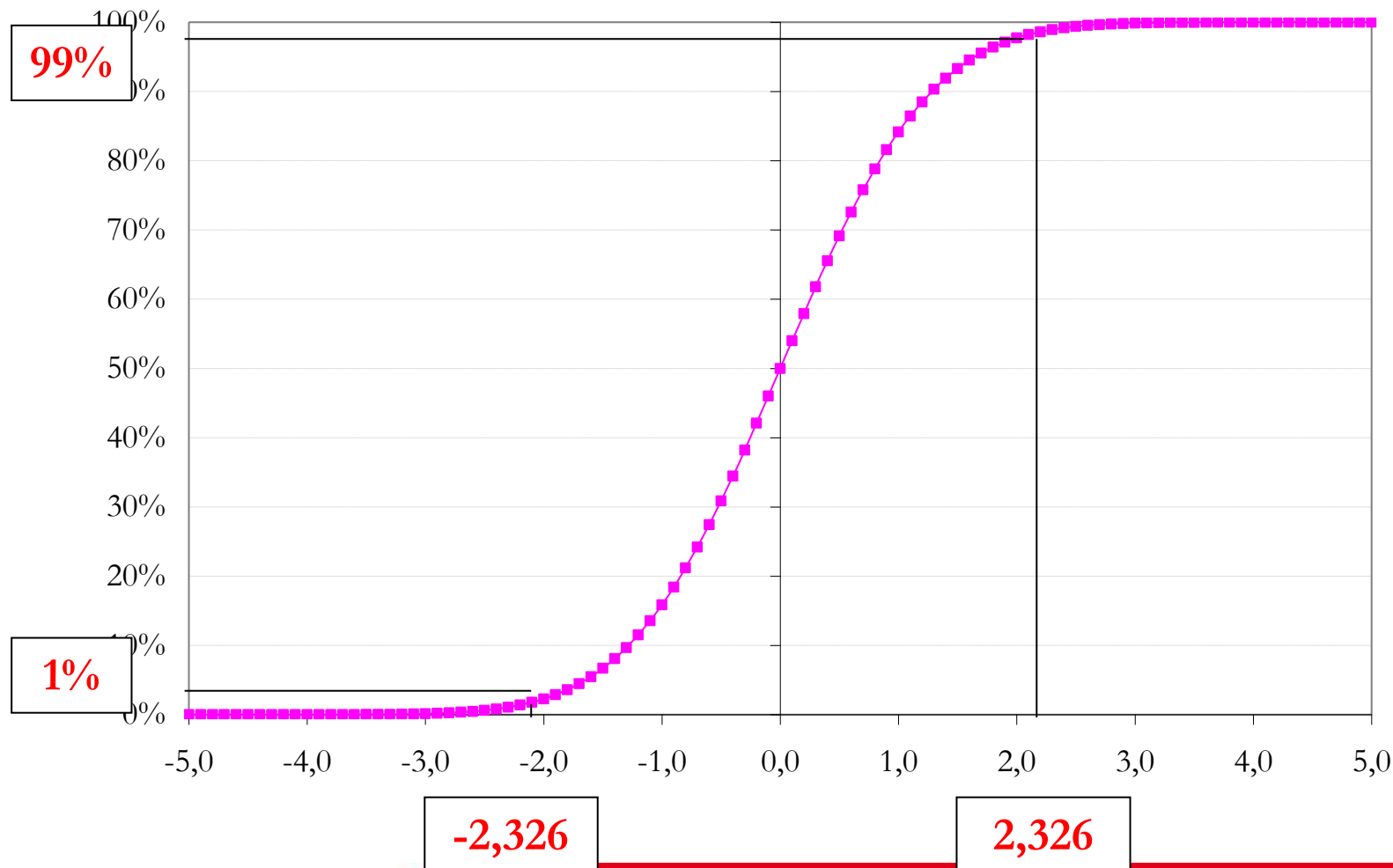
Distribution function of normal distribution

95% one-sided confidence interval



Distribution function of normal distribution

○ 99% one-sided confidence interval



Expected shortfall

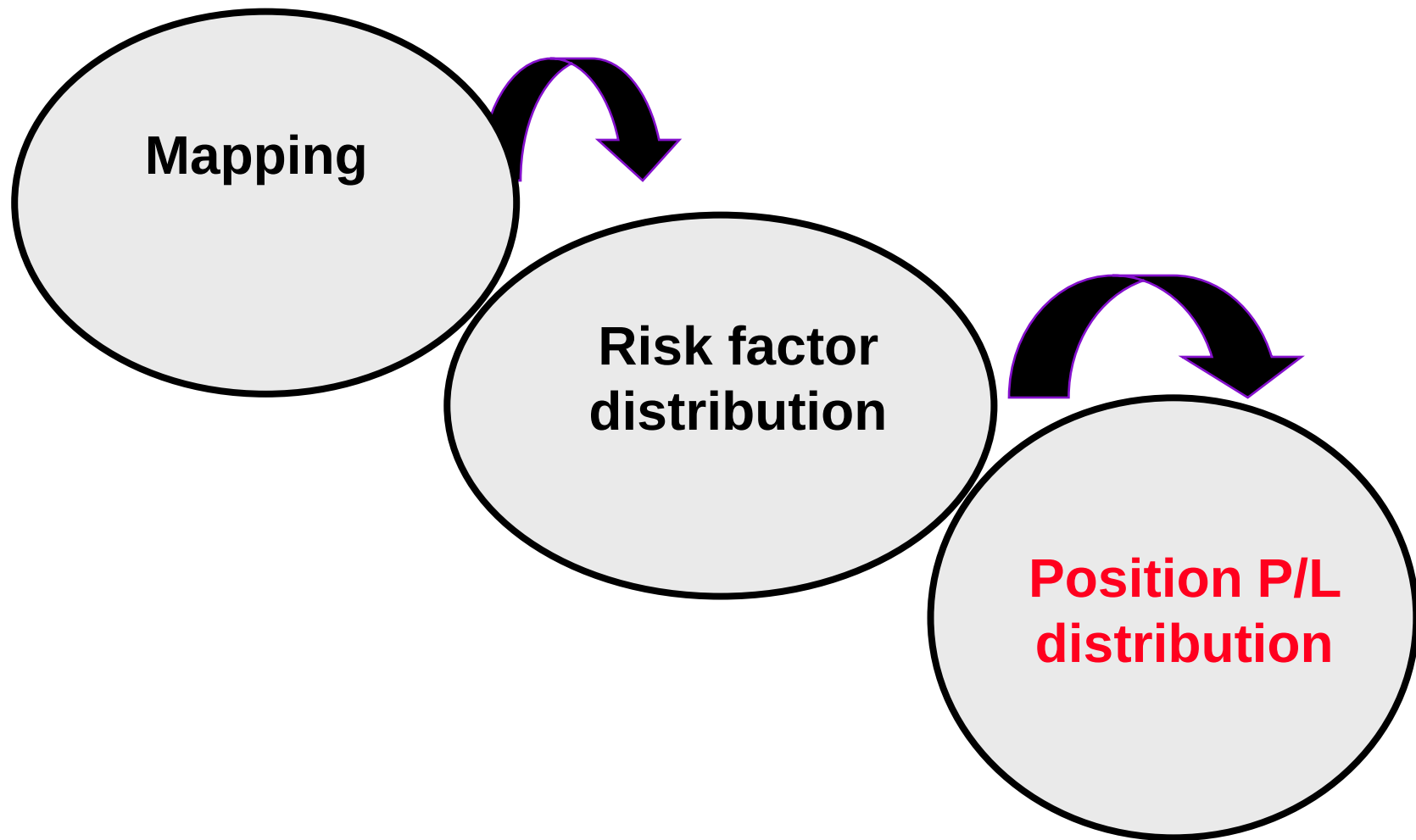
- The expected value of the loss in a specified worst-case fraction of cases (confidence level), over a given time period and under normal conditions.
- Shows the expected value of losses in excess of VaR in a continuous scenario.

ES in the case
of a normal
distribution:

$$ES = \mu + \sigma \frac{e^{-Y^2 / 2}}{\sqrt{2\pi}(1 - \alpha)}$$

$$Y = N^{-1}(1 - \alpha)$$

Logical model for determining the Value at Risk



VaR easily, quickly - 3. step

delta normal VaR of a stock position of n different shares

- Weights for each risk factor (shares):
 - $(w_1; w_2; w_3 \dots w_n)$ where $w_i: V_i/V$, is the % of the total portfolio represented by the stock in question
- Assuming that the returns on each risk factor are normally distributed, then the portfolio return distribution is also normal and the VaR is calculated using the known formula for the **standard deviation of the portfolio return**:

$$VaR = \text{multiplier} \cdot \sigma_V = \boxed{\text{multiplier} \cdot \sqrt{(wCw)}} = \boxed{\text{multiplier} \cdot \sqrt{(w\sigma)R(w\sigma)}}$$

- C: covariance matrix of risk factor returns,
- R: correlation matrix of risk factor returns, σ_r the standard deviation vector of returns
- w: weights of each risk factor

The Value at Risk of a position depending on several risk factors:

*The VaR values for each risk factor add up if
and only if the risk factors are perfectly
correlated...*

*...i.e. **diversification** can also reduce the
maximum loss*

Portfolio VaR examples

- University student (in Sept 2021)
 - As a student, we have 23.000 HUF, 18.000 of which we can freely "invest" for 1 month, 5.000 we put aside in cash - but we want to pay our share of the rent, 20.000 HUF.
 - Is it an acceptable risk to buy OTP shares (1 share) with this money? And other shares?
- Fund manager in times of a crisis (in Sept 2012)
 - A fund manager has (had) a portfolio of shares:
 - 38 MOL: exchange rate at the time 18.100 Ft
 - 110 OTP: exchange rate at the time 3.969 Ft
 - 27 Richter: exchange rate at the time 39.000 Ft
 - How much risk are we facing? Is this risk acceptable if the fund manager does not want to lose more than 150.000 HUF in the short term (2 weeks)?
 - The fund manager calculates VaR from 2012 data...

Input data: exchange rates

date	MOL	OTP	Richter
2012.09.11	18 100	3 969	39 000
2012.09.10	17 460	3 919	38 950
2012.09.07	17 305	3 882	39 200
2012.09.06	16 800	3 779	39 450
2012.09.05	16 490	3 750	39 800
2012.09.04	16 550	3 699	39 690
2012.09.03	16 560	3 630	39 085
...			
2012.04.19	17 200	3 500	37 070
Units	38	110	27
Position	687 800	436 590	1 053 000

What is the next step ?

bankárképző

TANÁCSADÁS ÉS OKTATÁS

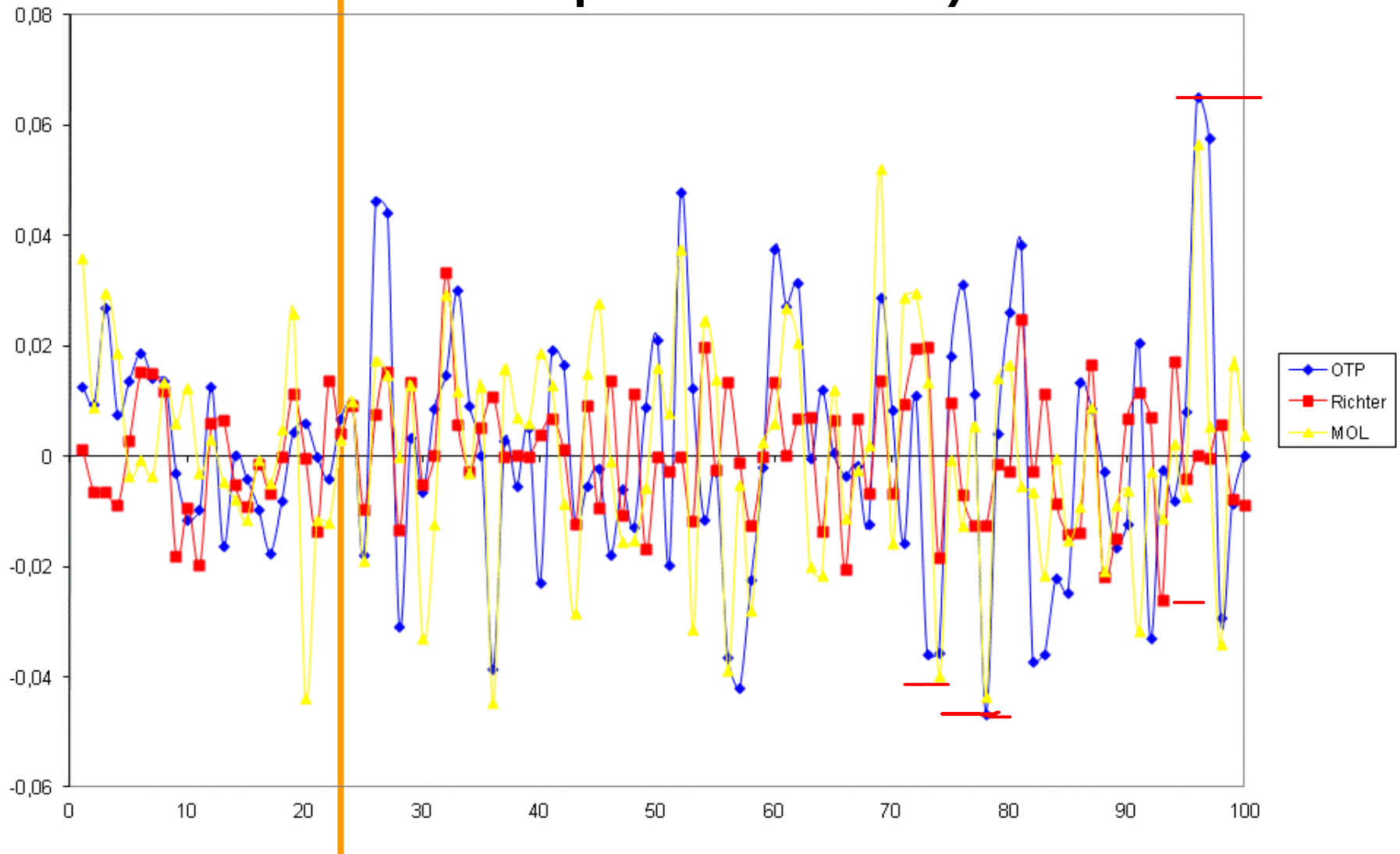


Calculation of logarithmic returns:

$$\ln(S_t) - \ln(S_{t-1})$$

	Logarithmic returns		
	MOL	OTP	Richter
1	3,6%	1,3%	0,1%
2	0,9%	0,9%	-0,6%
3	3,0%	2,7%	-0,6%
4	1,9%	0,8%	-0,9%
5	-0,4%	1,4%	0,3%
6	-0,1%	1,9%	1,5%
...			
100	0,4%	0,0%	-0,9%

The return standard deviations of OTP and MOL are "visibly" higher, so the negative returns are higher (and so are the positive returns)



Individual vs portfolio standard deviations and VAR

	MOL	OTP	Richter	Portfolio
Mean	0,05%	0,13%	0,04%	0,06%
Standard deviation	2,00%	2,20%	1,14%	1,21%

Correlation matrix	MOL	OTP	Richter
MOL	1,00	0,48	0,30
OTP	0,48	1,00	0,19
Richter	0,30	0,19	1,00
Position	687 800	436 590	1 053 000
Portfolio weights (w)	0,32	0,20	0,48

$$\sigma_V = \sqrt{(w\sigma)R(w\sigma)}$$

	MOL	OTP	Richter	Portfolio
VaR 95%, 1 day	3,3%	3,6%	1,9%	2,0%
VaR 95%, 10 days	10,4%	11,5%	5,9%	6,3%
VaR 99%, 1 day	4,6%	5,1%	2,7%	2,8%
VaR 99%, 10 days	14,7%	16,2%	8,4%	8,9%
VaR 99%, 20 days		22,9%		12,6%

VaR Ft	MOL	OTP	Richter	Portfolio	Without diversification	Diversification „gains”
VaR 95%, 1 day	22 575	15 821	19 784	43 410	58 179	14 769
VaR 99%, 10 days	100 967	70 758	88 481	194 152	260 206	66 054

VaR = Multiplier · Position size · Standard deviation

and deriving T-day VaR from 1-day VaR in the delta normal case:

$$T\text{-day VaR} = 1\text{-day VaR} \times \sqrt{T}$$

...but only if

- our method is delta normal
- we have 1-day returns
- we assume that the returns on consecutive days are independent

In all other cases T-day returns are calculated

Portfolio examples: conclusion

○ University student now

- If you use the % results of the previous VaR calculation, you can see that in 1 month (20 working days) you can lose more than 4.000 HUF on 1 OTP security (OTP exchange rate 2021.09.23: 17895 HUF, $\text{Var (99\%, 20 days)} = 17895 \times 22,9\% = 4098$ HUF). Thus, investing only in OTP is too risky, as you may have less than the 20.000 HUF invested and needed at the end of the month, after 1 month if such a "maximum loss" occurs $((17895 - 4098) + 5000 = 18797 < 20000$.
- It would also be better to invest in a more diversified 3-element equity portfolio with the same composition as in the example, because its VaR is much lower (12.6% for 20 days, which is only 2308 HUF $(17895 \times 12.9\%)$ for the same investment), which does not jeopardize the payment of the rent (even with the VaR loss, we will still have more money than 20.000 $(17895 - 2308 + 5000 = 20587 > 20000)$...
- ...but for such a short term, it is better to invest in something with a lower yield but even lower risk (e.g.: a bond)

○ Fund Manager

- Even though the 1-day 95% VaR is only 43.410 HUF, the chance of "error" is too high, and if we look ahead 10 days with 99% confidence ("safety"), we can lose more than 150.000 HUF
- The portfolio needs to be restructured by selling some of the shares and buying lower risk assets

Thank you for your attention!